Practical – 05

Part – A

Shape areas

AIM :

To implement an abstract `Shape` class with an abstract `area()` method and demonstrate **inheritance, method overriding, runtime polymorphism, and a mixed** `Shape[]` **array** using `Circle`, `Rectangle`, and `Triangle`, while calculating the **area, total area, and largest area**.

CODE :

25IT033 > Main.java > ...

```
50  public class Main {
51      public static void main(String[] args) {
68              System.out.print("Enter your choice: ");
69              int choice = sc.nextInt();
70
71              switch (choice) {
72
73                  case 1:
74                      System.out.print("Enter radius: ");
75                      double radius = sc.nextDouble();
76                      shapes[i] = new Circle(radius);
77                      break;
78
79                  case 2:
80                      System.out.print("Enter length: ");
81                      double length = sc.nextDouble();
82
83                      System.out.print("Enter width: ");
84                      double width = sc.nextDouble();
85
86                      shapes[i] = new Rectangle(length, width);
87                      break;
88
89                  case 3:
90                      System.out.print("Enter base: ");
91                      double base = sc.nextDouble();
92
93                      System.out.print("Enter height: ");
94                      double height = sc.nextDouble();
95
96                      shapes[i] = new Triangle(base, height);
97                      break;
98
99                  default:
100                     System.out.println("Invalid choice!");
101                     i--; // Ask again for the same shape
```

25IT033 > Main.java > ...

```
34
35  class Triangle extends Shape {
36      private double base;
37      private double height;
38
39      Triangle(double base, double height) {
40          this.base = base;
41          this.height = height;
42      }
43
44      @Override
45      double area() {
46          return 0.5 * base * height;
47      }
48  }
49
50  public class Main {
        Run main | Debug main
51      public static void main(String[] args) {
52
53          Scanner sc = new Scanner(System.in);
54
55          System.out.print("Enter number of shapes: ");
56          int n = sc.nextInt();
57
58          Shape[] shapes = new Shape[n];
59
60          // Take input from user
61          for (int i = 0; i < n; i++) {
62
63              System.out.println("\nShape " + (i + 1));
64              System.out.println("1. Circle");
65              System.out.println("2. Rectangle");
66              System.out.println("3. Triangle");
67
68              System.out.print("Enter your choice: ");
```

```
25IT033 > J Main.java > ...
 50    public class Main {
 51        public static void main(String[] args) {
101                        i--; // Ask again for the same shape
102                    }
103                }
104
105                double total = 0;
106                double largest = 0;
107                Shape largestShape = null;
108
109                // One loop for all shapes
110                for (Shape shape : shapes) {
111
112                    double currentArea = shape.area();
113
114                    System.out.println("Area: " + currentArea);
115
116                    total += currentArea;
117
118                    if (currentArea > largest) {
119                        largest = currentArea;
120                        largestShape = shape;
121                    }
122                }
123
124                System.out.println("\nTotal Area: " + total);
125                System.out.println("Largest Area: " + largest);
126
127                sc.close();
128            }
129        }
130
```

OUTPUT :

```
Enter number of shapes:
3


Shape 1
1. Circle
2. Rectangle
3. Triangle
Enter your choice:
1

Enter radius:
6


Shape 2
1. Circle
2. Rectangle
3. Triangle
Enter your choice:
2

Enter length:
5

Enter width:
8


Shape 3
1. Circle
2. Rectangle
3. Triangle
Enter your choice:
3

Enter base:
5
```

```
Enter base:
5

Enter height:
8

Area: 113.09733552923255
Area: 40.0
Area: 20.0

Total Area: 173.09733552923257
Largest Area: 113.09733552923255
```

Learning Outcome :

Understand abstract classes and abstract methods in Java.

Learn how to use inheritance (extends) to create different shape types.

Understand method overriding and runtime polymorphism.

Learn how a parent-type Shape[] can store different child objects.

Practice processing different objects using one loop through polymorphism.

Learn to calculate individual areas, the total area, and the largest area.

Payroll

Aim

To implement an abstract Employee class with an abstract monthlySalary() method and demonstrate inheritance, method overriding, runtime polymorphism, super, and instanceof using FullTime, PartTime, and Intern employee types, while calculating individual salaries and the total payroll.

CODE:

25IT033 > Main.java > Main > main(String[] args)

```java
abstract class Employee {
    protected String name;
    protected int id;

    Employee(String name, int id) {
        this.name = name;
        this.id = id;
    }

    abstract double monthlySalary();
}

class FullTime extends Employee {
    private double salary;

    FullTime(String name, int id, double salary) {
        super(name, id);
        this.salary = salary;
    }

    @Override
    double monthlySalary() {
        return salary;
    }
}

class PartTime extends Employee {
    private double hours;
    private double rate;

    PartTime(String name, int id, double hours, double rate) {
        super(name, id);
        this.hours = hours;
        this.rate = rate;
    }
```

25IT033 > Main.java > Main > main(String[] args)

```java
class PartTime extends Employee {

    @Override
    double monthlySalary() {
        return hours * rate;
    }
}

class Intern extends Employee {
    private double stipend;

    Intern(String name, int id, double stipend) {
        super(name, id);
        this.stipend = stipend;
    }

    @Override
    double monthlySalary() {
        return stipend;
    }
}

public class Main {
    Run main | Debug main
    public static void main(String[] args) {

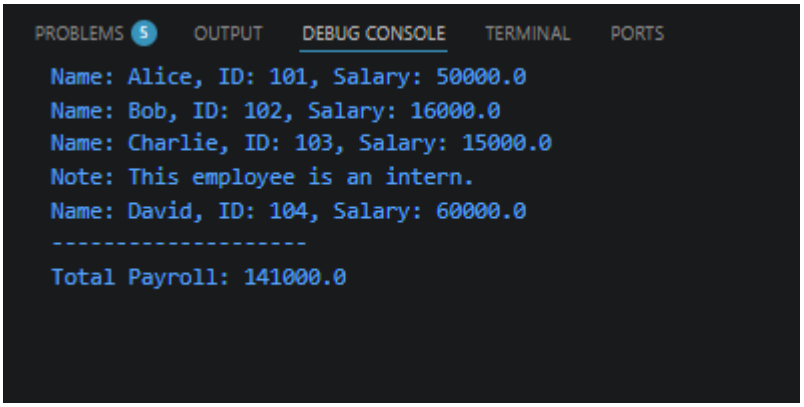
        Employee[] employees = {
            new FullTime("Alice", 101, 50000),
            new PartTime("Bob", 102, 80, 200),
            new Intern("Charlie", 103, 15000),
            new FullTime("David", 104, 60000)
        };

        double total = 0;

        for (Employee employee : employees) {
```

```
69          for (Employee employee : employees) {
70
71              double salary = employee.monthlySalary();
72
73              System.out.println(
74                  "Name: " + employee.name +
75                  ", ID: " + employee.id +
76                  ", Salary: " + salary
77              );
78
79              if (employee instanceof Intern) {
80                  System.out.println("Note: This employee is an intern.");
81              }
82
83              total += salary;
84          }
85
86          System.out.println("--------------------");
87          System.out.println("Total Payroll: " + total);
88      }
89  }
90
```

OUTPUT :



Learning Outcome :

Understand abstract classes and abstract methods in Java.

Learn to implement inheritance using extends.

Understand method overriding for different employee types.

Learn to use super to initialize common employee details.

Understand runtime polymorphism using an Employee[] array.

Learn to use instanceof to identify specific employee types.

Calculate individual salaries and the total monthly payroll.